

Snowflake Cloud Data Warehouse Cheat Sheet

Course: HEIG-VD IST 2025/26

Topic: Snowflake Cloud Data Warehouse

Authors: Markaj Agron · Jorand Yuuta · Stampfli Nathan · Liao Pei-Wen

Date: June 2026

Why Snowflake?

Real-world scenario: SwissBike SA

SwissBike SA is a Swiss SME selling bicycles and accessories online. As the company grows, data is generated by multiple systems:

- E-commerce platform
- ERP system
- CRM system
- Marketing tools

Management needs reliable answers to questions such as:

- Which countries generate the most revenue?
- Who are our most valuable customers?
- Which products should we promote?
- How are sales evolving over time?

Without a centralized solution, data is scattered across multiple systems and reports become slow, inconsistent and difficult to maintain.

What does Snowflake do?

Snowflake is a cloud-native data warehouse that centralizes data from multiple sources and enables large-scale analytical processing.

Data Storage Characteristics

- Supports structured, semi-structured and unstructured data
- Native support for JSON, Avro, ORC, Parquet and XML
- Automatic compression
- Micro-partitioning for storage and query optimization
- Separation of Storage and Compute

Inputs

- CSV files
- Databases
- ERP systems
- CRM systems
- APIs
- Streaming data

Outputs

- SQL query results
- Dashboards (Power BI, Tableau, Looker)
- Reports
- Data for machine learning and analytics

Benefits and limitations

Benefits

- Separation of Storage and Compute: Storage and compute resources can be scaled independently, improving flexibility and cost management.
- Broad Data Support: Supports structured, semi-structured and unstructured data, with native support for JSON, Avro, ORC, Parquet and XML.
- High Scalability: Compute resources are organized into Virtual Warehouses that can scale up (larger warehouses) or scale out (multi-cluster warehouses).
- Excellent Concurrency: Multiple Virtual Warehouses can access the same data simultaneously with minimal resource contention.
- Massively Parallel Processing (MPP): Queries are distributed across multiple compute resources, enabling efficient processing of large datasets.
- Minimal Administration: Fully managed service with no infrastructure maintenance, indexing or vacuuming required.
- Strong Security: Encryption at rest and in transit, role-based access control (RBAC), MFA and network policies.
- Time Travel & Fail-safe: Supports recovery of deleted or modified data and provides access to historical versions of data.
- Fast Analytical Performance: Query optimization, result caching and micro-partitioning improve analytical query performance.
- Secure Data Sharing: Share live data securely across accounts, organizations and cloud providers without data duplication.
- Multi-cloud Support: Available on AWS, Microsoft Azure and Google Cloud Platform.
- Automatic Cost Optimization: Warehouses can automatically suspend when idle and resume on demand.

Limitations

- Cost: Can become expensive for large workloads, poorly optimized queries or oversized warehouses.
- Unpredictable Spending: Consumption-based pricing can make costs difficult to estimate and control.
- Vendor Lock-in: Migrating data pipelines, security policies and workloads to another platform may require significant effort.
- Cloud-only Platform: No on-premises deployment option.
- Limited Infrastructure Control: Less flexibility than self-managed solutions for low-level tuning and customization.
- Time Travel Storage Costs: Retaining historical data increases storage consumption.
- Learning Curve: Understanding warehouses, credits, clustering, optimization and cost management requires training.
- Not Optimized for OLTP: Designed for analytical workloads (OLAP), not high-frequency transactional applications.
- Dependency on Cloud Providers: Availability depends on Snowflake and the underlying cloud provider infrastructure.

Cost structure

Snowflake pricing is mainly based on:

1. Storage

Measured in:

- Terabytes (TB) stored per month

Example:

1 TB stored data

Cost: approximately \$23/month

2. Compute

Measured in:

- Credits consumed by Virtual Warehouses

Factors:

- Warehouse size
- Execution time
- Number of clusters

Monthly cost example

SwissBike SA

Assumptions:

- 1 TB of stored data
- 10 business users
- Approximately 10 hours of warehouse usage per week
- X-Small warehouse
- Snowflake AI features enabled (CoWork and CoCo)

Component	Estimated Cost
Compute (Warehouse)	\$117.00/month
Snowflake CoWork	\$123.08/month
Snowflake CoCo	\$5.78/month
Storage (1 TB)	\$23.00/month
Total	\$268.86/month

Cost observations

- Compute is the largest cost component.
- AI features (CoWork and CoCo) can significantly increase the monthly bill.
- Storage costs are relatively low compared to compute costs.
- Automatic warehouse suspension can help reduce compute expenses.

How to get started

Prerequisites

- Snowflake account
- Web browser
- Basic SQL knowledge

Create a database

```
CREATE DATABASE POC_SNOWFLAKE;
```

Create a schema

```
CREATE SCHEMA DEMO;
```

Create a warehouse

```
CREATE WAREHOUSE DEMO_WH  
WITH WAREHOUSE_SIZE = 'XSMALL';
```

Verify the environment

```
SELECT  
    CURRENT_ACCOUNT(),  
    CURRENT_USER(),  
    CURRENT_WAREHOUSE();
```

Hello World example

```
SELECT *  
FROM SNOWFLAKE_SAMPLE_DATA.TPCH_SF1.CUSTOMER  
LIMIT 10;
```

Common commands / operations / usage patterns

Query sample data

```
SELECT *  
FROM SNOWFLAKE_SAMPLE_DATA.TPCH_SF1.ORDERS  
LIMIT 100;
```

Count records

```
SELECT COUNT(*)  
FROM CUSTOMER;
```

Revenue by country

```
SELECT  
    N.N_NAME,  
    SUM(L.L_EXTENDEDPRICE) AS REVENUE  
FROM LINEITEM L  
JOIN ORDERS O  
    ON L.L_ORDERKEY = O.O_ORDERKEY  
JOIN CUSTOMER C  
    ON O.O_CUSTKEY = C.C_CUSTKEY  
JOIN NATION N  
    ON C.C_NATIONKEY = N.N_NATIONKEY  
GROUP BY N.N_NAME;
```

Resize a warehouse

```
ALTER WAREHOUSE COMPUTE_WH  
SET WAREHOUSE_SIZE='SMALL';
```

Suspend a warehouse

```
ALTER WAREHOUSE COMPUTE_WH  
SUSPEND;
```

Resume a warehouse

```
ALTER WAREHOUSE COMPUTE_WH  
RESUME;
```

Time Travel

```
DROP TABLE CUSTOMER_PROD;
```

```
UNDROP TABLE CUSTOMER_PROD;
```

- Time Travel: recover historical versions of data
- Fail-safe: additional recovery period managed by Snowflake

Zero-Copy Clone

```
CREATE TABLE CUSTOMER_TEST  
CLONE CUSTOMER_PROD;
```

Query Profile (MPP)

After executing a query:

Monitoring

- Query History
- Query Profile

The Query Profile displays:

- Table Scan
- Join
- Aggregate
- Sort

and helps understand how Snowflake executes analytical workloads using MPP.

When should a SME use Snowflake?

Recommended

- Business Intelligence
- Data Warehousing
- Analytics
- Reporting
- Data Science
- Multi-source data integration

Not recommended

- Small applications with limited data
- OLTP workloads
- Projects requiring full infrastructure control

Recommendation: Snowflake is an excellent choice for SMEs that need scalable analytics with minimal operational overhead, but organizations must actively monitor costs and evaluate vendor lock-in risks.